RAMCO INSTITUTE OF TECHNOLOGY

Department of Computer Science and Engineering

Academic Year: 2019 - 2020 (Even Semester)

Degree, Semester & Branch: IV Semester B.E. Computer Science and Engineering

Course Code & Title: CS8451 & Design and Analysis of Algorithms

Name of the Faculty member: Ms.P.Jothi Thilaga, AP/CSE

Date & Venue: 08.01.2020 & CRXXIII

Innovative Practice: Minute Paper

Topic: Asymptotic Notations

<u>Type of Learning:</u>

Active Learning

<u>Learning Objectives:</u>

O1: To make the students to understand the concept of asymptotic notation and its properties.

<u>Description:</u>

Minute paper is a short writing activity within the classroom in which students generate answers in response to questions asked by the teacher regarding the concepts taught in class. These questions stimulate a student to reflect on the lesson taught and learned and provides feedback regarding their understanding which helps the teachers to plan ahead for the next class. Its major advantage is that it provides rapid feedback on whether the instructor's main idea and what the students perceived as the main idea are the same.

<u>Uses of Minute Paper:</u>

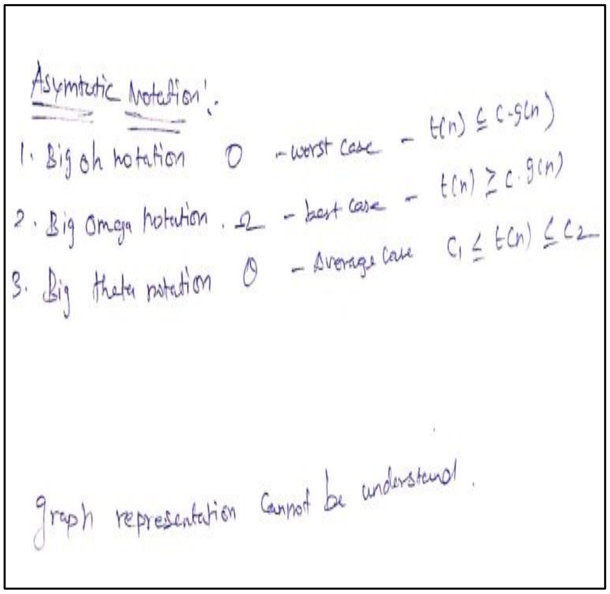
- This activity provides a conceptual bridge between successive class periods.
- Improve the quality of class discussion by having students write briefly about a concept or issue before they begin discussing it.

<u>Justification for chosen the topic:</u>

Asymptotic notation is an important topic in Design and Analysis of algorithms. These notations are used in the analysis part of the algorithms. The algorithm’s efficiency is analyzed and representing using these notations. This activity makes the students to get a sound knowledge in this concept.

Implementation of Minute Paper:

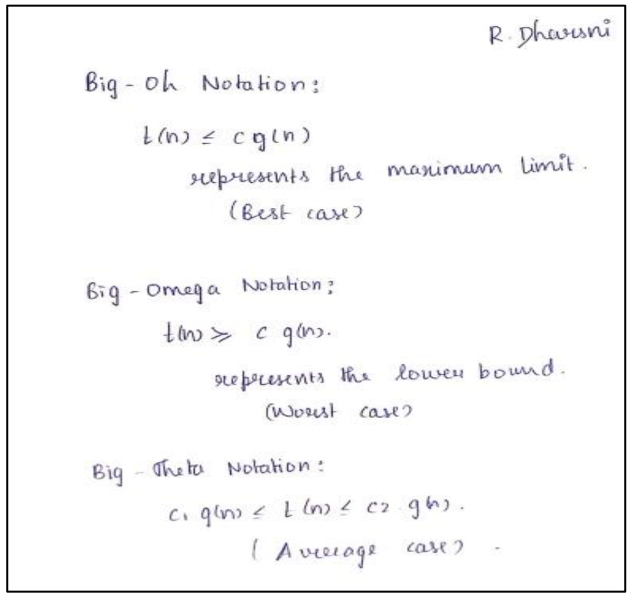
At the end of the class, students were asked to write about the topic discussed in the class. The students expressed the understood content and the content which were not clear in that particular topic. The students also write about the specific topic which is needed to be discussed or clarified in the further classes. This activity shows whether the students can able to understand the specific topic and their involvement the particular class.



Asymtotic Notation:

1. Big oh notation O - worst case - $t(n) \le c \cdot g(n)$
2. Big Omega notation. Ω - best case - $t(n) \ge c \cdot g(n)$
3. Big theta notation Θ - Average case $c_1 \le t(n) \le c_2$

graph representation cannot be understood.



R. Dharsni

Big - oh Notation:

$t(n) \le c\,g(n)$

represents the maximum limit.

(Best case)

Big - Omega Notation:

$t(n) \geq c\,g(n)$.

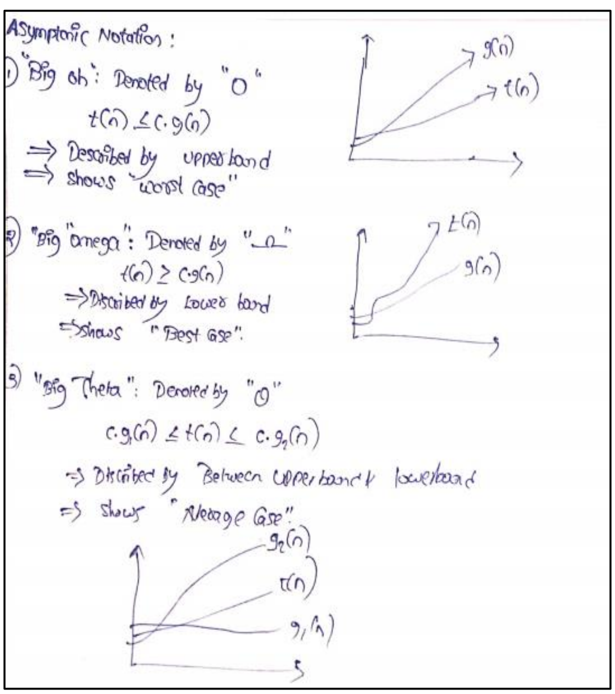
represents the lower bound.

(Worst case)

Big - Theta Notation:

$c_1\,g(n) \le t(n) \le c_2\,g(n)$.

(Average case)



Asymptotic Notation:

1) "Big oh": Denoted by "O"

$t(n) \le c \cdot g(n)$

⇒ Described by upper bound

⇒ Shows "worst case"

2) "Big omega": Denoted by "Ω"

$t(n) \ge c\,g(n)$

⇒ Described by Lower bound

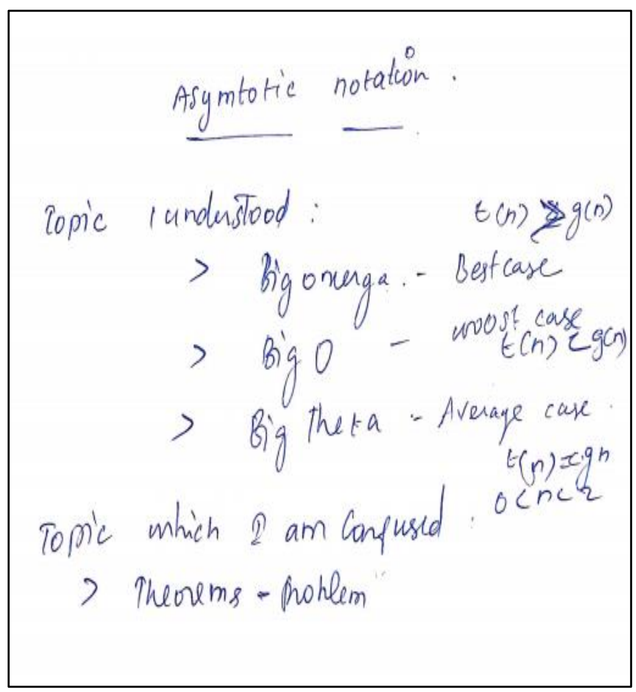
⇒ Shows "Best case".

3) "Big Theta": Denoted by "Θ"

$c \cdot g_1(n) \le t(n) \le c \cdot g_2(n)$

⇒ Described by Between upper bound & lower bound

⇒ Shows "Average case".



Asymtotic notation.

Topic I understood: $t(n) \geq g(n)$

> Big omega. - Best case

> Big O - worst case $t(n) \le g(n)$

> Big theta - Average case. $t(n) = c\,g\,n$ $0 < n < 2$

Topic which I am confused

> Theorems - problem

Outcomes:

- Use of minute papers in the classroom actually increases attentiveness in the class and also improves the students' skill of writing and critical analysis of the topics.
- This activity reflects the understanding and involvement of the students in the particular topic.

Relevance to POs

Objective	PO1	PO2	PO3	PO4	PO5	PO9	PO12
O1	3	3	2	2	2	3	2

Reflective Report:

Identified Problems

- Some students are hesitated and lack to share the concepts that they learnt in the class.

Initiatives to address the problems

- Make the students to know the impact and importance of sharing their views and understanding related to the topic and made them involve in the activity.
- Addressed the students that the activity is not a kind of evaluation methodology to assess.

Post-implementation

- Students were actively participated in this activity.
- From this activity, the muddiest parts in the concept were identified and the properties of asymptotic notations were explained in the next class for well understanding.
- The graphical representations of the asymptotic notation were again explained for their better understanding.

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